1.Introduction

In this report, I investigate a one-dimensional regression problem where the goal is to approximate an unknown function from noisy observations. The dataset consists of input-output pairs (x, y), where the outputs are generated from an underlying function with added noise.

Two different modelling approaches are considered:

- A fourth-order polynomial regression model using least squares
- A feedforward neural network model

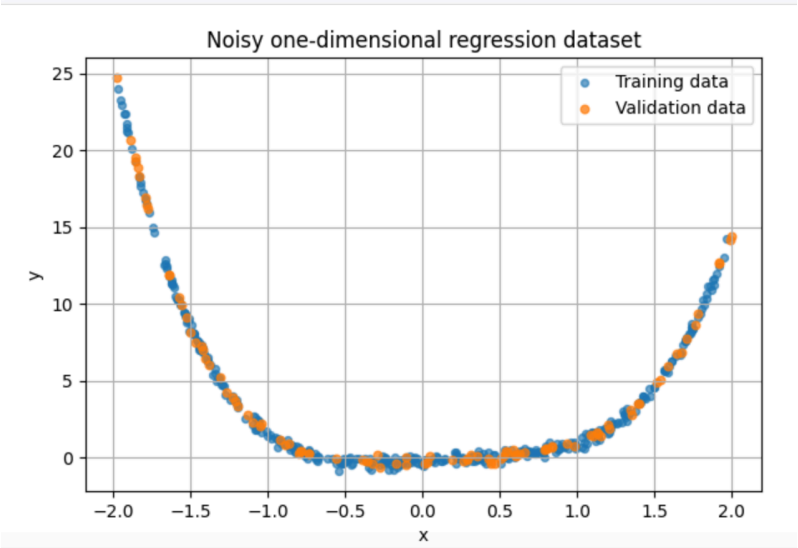
The objective is not only to compare predictive performance but also to analyse differences in optimisation behaviour, model flexibility, and generalisation ability.

2. Dataset Description

The dataset is divided into a training set and a validation set:

- Training set size: 400 samples
- Validation set size: 100 samples

Each sample contains a one-dimensional input x and a corresponding output y.



From the plot, the following observations can be made:

The relationship between x and y is clearly nonlinear, with a smooth symmetric structure resembling a low-degree polynomial function. This suggests that a polynomial model may already provide a suitable approximation for the underlying function. There is visible noise in the data, especially near the centre

This further confirms that both simple parametric models and more flexible models could be appropriate for this task.

3.Method 1: Least Squares Polynomial Regression

3.1 Model Description

A fourth-order polynomial model is used:

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4$$

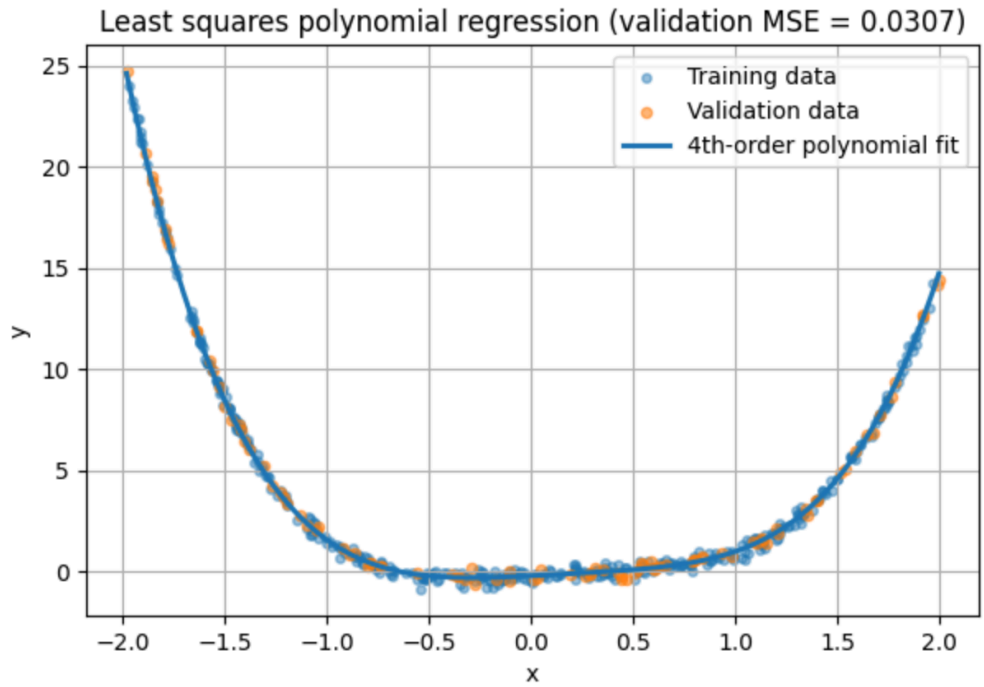
3.2 Results

The fitted coefficients are:

- $a_0 = -0.2146$
- $a_1 = 0.5076$
- $a_2 = 0.2899$
- $a_3 = -0.8037$
- $a_4 = 1.2054$

The performance of the model is:

- Training MSE: **0.042250**
- Validation MSE: **0.030731**



3.3 Discussion

The polynomial model provides a very good fit to the data. The predicted curve closely follows the overall trend of the observations, especially capturing the smooth curvature of the dataset.

An important observation is that the validation MSE is slightly lower than the training MSE. This indicates that the model generalises well and does not overfit the training data. This behaviour is expected because the polynomial model has limited flexibility and cannot easily fit noise.

This behaviour can be explained by the bias-variance trade-off. The polynomial model has relatively low variance due to its limited flexibility, which helps prevent overfitting in the presence of noise.

4. Method 2: Neural Network Regression

4.1 Model Description

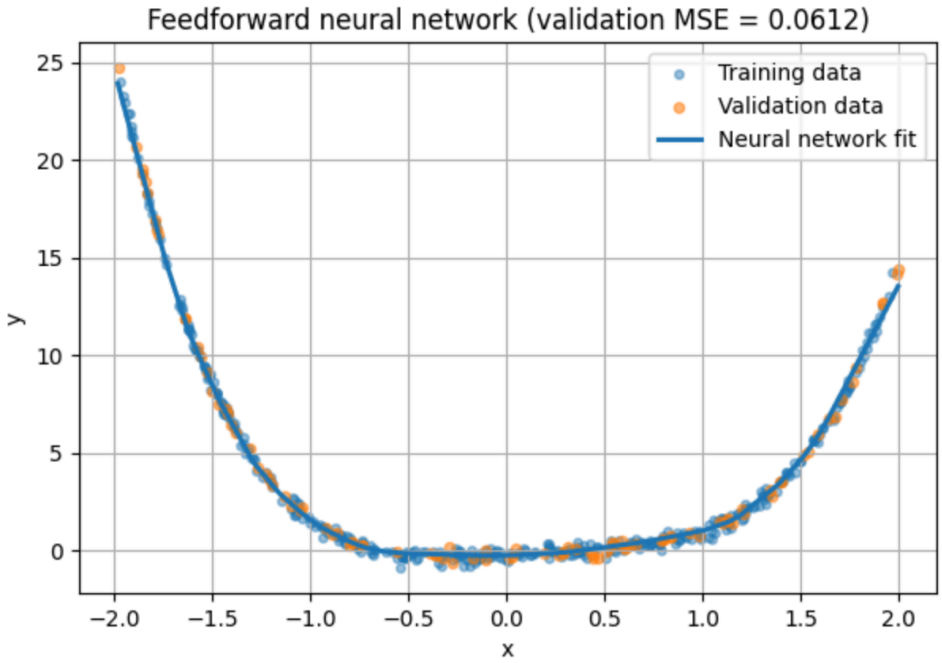
A feedforward neural network was used with the following configuration:

- Hidden layers: (64, 64)
- Activation function: ReLU
- Optimiser: Adam (default in MLPRegressor)
- Regularisation parameter: \alpha = 1 x 10^{-5}

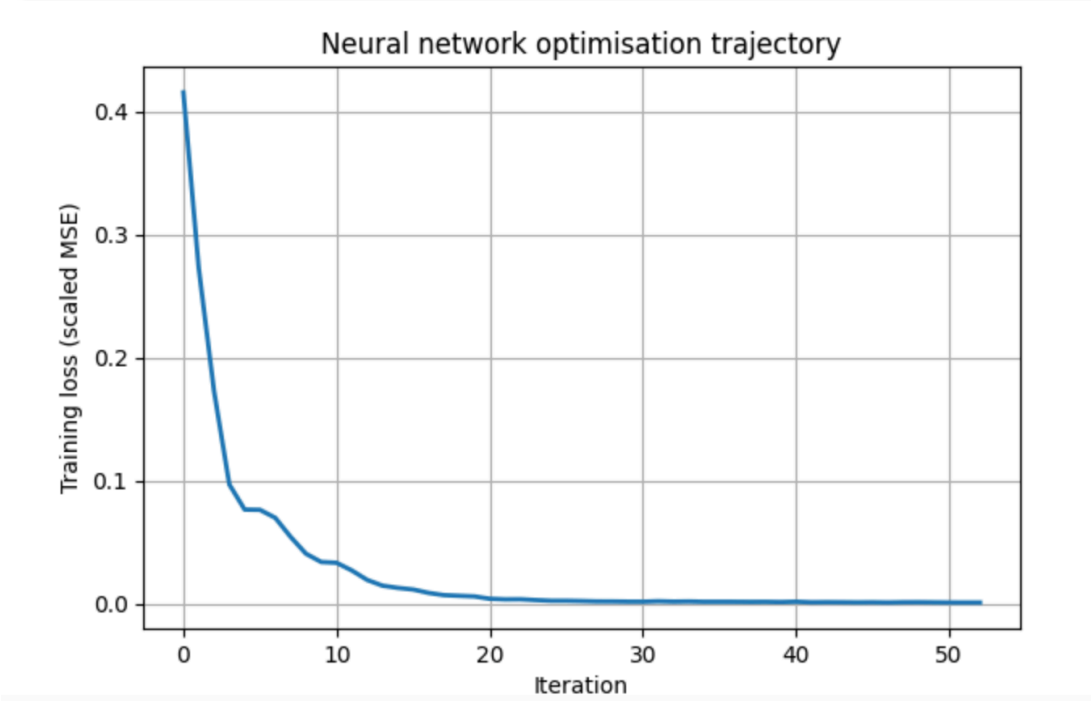
4.2 Results

The best-performing neural network achieved:

- Training MSE: **0.055317**
- Validation MSE: **0.061199**



4.3 Training Behaviour



The training loss decreases rapidly at the beginning and then gradually converges. This indicates that the optimisation process is stable and effective.

However, despite good convergence, the final validation error remains higher than the polynomial model.

4.4 Discussion

The neural network is capable of capturing the general nonlinear structure of the data. However, compared to the polynomial model, its performance is worse.

Several reasons can explain this:

- **Model complexity vs data size: The neural network has significantly more parameters than the polynomial model, making it more prone to overfitting.**
- **Noise sensitivity: Neural networks tend to fit noise more easily due to their flexibility.**
- **Optimisation limitations: Gradient-based methods may converge to local minima.**
- **Model mismatch: The underlying function appears close to a polynomial.**

Overall, these factors explain why the neural network, despite its flexibility, does not outperform the simpler polynomial model in this specific setting.

5. Comparison of Methods

5.1 Performance Comparison

Model	Training MSE	Validation MSE
Polynomial Regression	0.042250	**0.030731**
Neural Network	0.055317	0.061199

The polynomial model consistently outperforms the neural network on both training and validation sets, indicating not only better fit but also stronger generalisation ability for this specific dataset.

5.2 Key Differences

Optimisation:

- Polynomial regression: closed-form solution, deterministic
- Neural network: iterative optimisation, dependent on hyperparameters

Model flexibility:

- Polynomial: limited but stable
- Neural network: highly flexible but sensitive

Generalisation:

- Polynomial: better generalisation in this task
- Neural network: higher risk of overfitting

5.3 Key Insight

A key insight from this experiment is that **a more complex model does not necessarily lead to better performance**. Although neural networks are more powerful in theory, they may underperform when:

- The dataset is small
- The underlying function is relatively simple
- Noise is present

In such cases, simpler models like polynomial regression can achieve better generalisation.

6. Conclusion

In this coursework, two different approaches were used to approximate a noisy one-dimensional function.

The fourth-order polynomial regression model provided strong performance with low validation error and good generalisation. The neural network, although more flexible, did not outperform the polynomial model due to its sensitivity to noise and higher model complexity.

This study highlights the importance of choosing an appropriate model based on the structure of the data rather than assuming that more complex models will always perform better.

7. Limitations and Future Work

Although the results obtained in this coursework are satisfactory, there are several limitations that should be acknowledged.

First, the neural network model was only explored with a limited range of architectures and hyperparameters. While a grid of hidden layer sizes and regularisation strengths was tested, more extensive tuning could potentially improve performance. For example, alternative architectures such as deeper networks or different activation functions may lead to better approximation of the underlying function.

Second, the dataset used in this task is relatively small, consisting of only 400 training samples. Neural networks typically require larger datasets to fully exploit their flexibility and generalisation capabilities. The limited data size may have restricted the ability of the neural network to outperform the polynomial model.

Third, the optimisation process for the neural network relies on gradient-based methods, which may converge to local minima rather than the global optimum. Although the training loss shows stable convergence, there is no guarantee that the final model is optimal. In contrast, the least squares method provides a closed-form solution, ensuring optimality under the given model assumptions.

Another limitation is that the models were evaluated using only a single train-validation split. This may introduce variability in the results depending on how the data is partitioned. A more robust evaluation approach, such as k-fold cross-validation, could provide a more reliable estimate of model performance.

Future Work

Several improvements could be explored in future work.

One possible direction is to apply **cross-validation** to obtain a more reliable comparison between models and reduce the effect of data splitting.

Another improvement would be to experiment with **regularisation techniques** in the neural network, such as dropout or stronger weight decay, which could help reduce overfitting and improve generalisation.

In addition, **feature transformations** or alternative basis functions (e.g., radial basis functions or spline models) could be explored as an extension of polynomial regression, potentially providing a balance between flexibility and interpretability.

Finally, increasing the size of the dataset or reducing noise could significantly improve the performance of the neural network, allowing it to better demonstrate its advantages over simpler models.